

Semiconductor Back-End 6D Digital Twin Simulator

Instructor Script Guide

7 Sessions • 60–120 min each • IEG Training Series
IEG Co., Ltd. — Global Business Division

I. Program Overview

This guide provides session-by-session scripts for teaching the five back-end stages (Die Bonding → Wire Bonding → Molding → Saw/Dicing → Package Testing) using IEG's 6D Digital Twin Simulator.

Defining the '6D Digital Twin'

- 3D — a realistic spatial model of actual equipment, dies, and packages
- 4D — reflects process progression along a time axis (cure cycles, bonding sequence, etc.)
- 5D — a data/intelligence layer where AI predicts and warns of anomalies from accumulated process data (the dashboard's five indicators)
- 6D — a spatial pattern intelligence layer that reads the package map's spatial pattern (Edge · Center · Donut · Scratch · Random · Local Cluster) at Package Testing to narrow down root-cause candidates

This Guide's Approach to Fault Cases

Every module session in this guide pairs its numeric Process Anomaly Scenario with a 6D Fault Case: a prediction of what spatial pattern that specific fault would produce on the final package map, and why. Two modules (Wire Bonding and Saw/Dicing) are deliberately given the same expected pattern — Scratch — for two different physical reasons, so learners practice the key 6D lesson: shape narrows the search, but does not replace checking a station's own data.

Session Overview

Session	Topic	Key Simulator Module	Duration
0	Orientation · 6D Digital Twin Overview (Back-End)	Full Dashboard / 3D View Navigation	60 min
1	Die Bonding	MODULE 01 · Collet Pick & Place / Cure	90 min
2	Wire Bonding	MODULE 02 · EFO · Loop · Vision Alignment	90 min
3	Molding	MODULE 03 · EMC · PID Control · Six Sigma Analysis	90 min
4	Saw / Dicing	MODULE 04 · SFC Design · EtherCAT · Reject Logic	90 min
5	Package Testing	MODULE 05 · AOI · X-ray · Bond Test · SPC	90 min
6	Integration · 5-Process Chain Scenario & Certification	Full Digital Twin Dashboard (fully linked)	120 min

** Each session runs in three stages: a 10-minute intro, 50–90 minutes of development/simulator practice, and 15–20 minutes of wrap-up and checkpoint review.*

II. [Session 0] Orientation · 6D Digital Twin Overview (Back-End)

Learning Objectives

- Understand the full flow of the five back-end modules (Die Bonding → Wire Bonding → Molding → Saw/Dicing → Package Testing).
- Explain how back-end's core purpose (mechanical assembly and reliability) differs from front-end.
- Understand the 6D digital twin concept (3D geometry + 4D time + 5D data/AI + 6D spatial pattern intelligence) in a back-end context.

Intro

Instructor: *Once a wafer finishes front-end testing, its individual dies are cut apart and carried through the five processes you'll learn starting today, ending as the chip inside your phone or your car.*

Instructor: *If front-end was a game of precisely controlling chemical and physical reactions, back-end is a game of mechanical assembly precision and reliability. We attach the die (Die Bonding), wire it up (Wire Bonding), encapsulate it (Molding), cut it apart (Dicing), and inspect it (Testing) — five stages.*

Development — Mapping the Five Back-End Processes

On the whiteboard, list the five processes in order and have learners summarize in one line what physically happens to the package at each stage.

Reviewing the 6D Digital Twin Definition

- 3D — realistic physical geometry of the equipment, dies, and packages
- 4D — process progression over time (cure cycles, bonding sequence, etc.)
- 5D — a data/AI layer that predicts and flags anomalies from the dashboard's five numeric indicators
- 6D — spatial pattern intelligence: reading the shape of failures on a package map at Package Testing (Edge · Center · Donut · Scratch · Random · Local Cluster) to narrow down root-cause candidates

Wrap-Up

Instructor: *The five-process map you drew today is your navigation for the next five weeks. Starting next session, we'll operate Die Bonding hands-on.*

Checkpoint Q&A

Q. How does back-end's core purpose differ from front-end?

A. Back-end physically assembles and packages the finished dies from a wafer into a mountable product, so mechanical reliability and assembly precision are the central concerns.

Q. Why does 6D matter especially in back-end?

A. The spatial pattern that appears on the final Package Testing map is a key clue for tracing back which of the five processes the problem actually started in.

II-1. [Session 1] Die Bonding

Learning Objectives

- Explain the principles and reliability-management points of the Die process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

Instructor: *Today is the first back-end stage: attaching an individual die to a leadframe or substrate with epoxy. Collet pick-and-place precision and cure management are central.*

Development — Simulator-Linked Instruction

- On the MODULE 01 screen, play the sequence where the collet vacuum-picks a die from the wafer and places it precisely onto the epoxy-dispensed leadframe.
- Explain how epoxy dispense volume and pick-and-place alignment error affect final reliability — under-dispense leaves voids beneath the die, and misalignment leads to downstream wire-bonding failures.
- Learner activity: adjust the Bond Force and cure-temperature sliders and observe how Defect Rate changes, recording results in the worksheet.

⚠ PROCESS ANOMALY SCENARIO — Void Formation from Under-Dispense

Root cause: A partially clogged dispenser nozzle drops the dispense volume below the 100% spec, leaving a void beneath the die.

Symptom (dashboard reaction): Defect Rate spikes sharply; Alignment Error also drifts out of spec at the same time.

Instructor action: Have learners readjust the dispense-volume and alignment sliders and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Local Cluster

A clogged dispenser nozzle degrades gradually, so a run of consecutively bonded units — not the whole tray — shares the defect before the drift is caught.

Instructor: *If this nozzle clog affected roughly 20 consecutive units before an operator caught it, what would you expect the final package map to look like? Not the whole tray — just a tight group.*

Contrast this with a fully random single-die defect: a clog-driven fault clusters in production sequence, while an isolated particle contamination event does not.

Wrap-Up

Instructor: *Today's Die process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between DAF and liquid epoxy?

A. DAF (Die Attach Film) offers highly uniform thickness; liquid epoxy is flexible but requires tighter dispense-volume control.

Q. What is the bottleneck of the Die Bonding process?

A. The cure oven's batch cycle time, which is difficult to parallelize and limits overall line throughput.

II-2. [Session 2] Wire Bonding

Learning Objectives

- Explain the principles and reliability-management points of the Wire process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

Instructor: *Today's process electrically connects die pads to the leadframe using fine gold or copper wire. EFO ball formation and capillary loop precision are the focus.*

Development — Simulator-Linked Instruction

- On the MODULE 02 screen, play the sequence where an EFO spark forms a ball at the wire tip, and the capillary traces an arched loop to the leadframe.
- Explain that too-low a loop height risks wire sag and shorting with neighboring wires, while too-high a loop risks wire sweep during molding.
- Learner activity: adjust EFO current and loop-height sliders and observe how ball size and loop shape change, comparing against the literature-validated optimal EFO current (~44 mA).

⚠ PROCESS ANOMALY SCENARIO — Loop Height Deviation Causing Wire Sag / Heel Crack

Root cause: Z-axis servo drift on the capillary pushes loop height outside the $150 \pm 15 \mu\text{m}$ spec, causing wire sag or heel cracking.

Symptom (dashboard reaction): Defect Rate rises; EFO current also drifts outside the $44 \pm 8 \text{ mA}$ spec at the same time.

Instructor action: Have learners readjust the loop-height and EFO-current sliders and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Scratch (Line)

Multi-head wire bonders assign fixed tray positions to each bonding head. If one specific head drifts out of spec while the others remain fine, every unit that head touched forms a straight line or row on the map — not because of physical contact damage, but because of a fixed equipment-to-position mapping.

Instructor: *This is one of the more advanced fault cases: a Scratch-shaped pattern normally means physical handling damage. But here, the exact same shape can come from a single bonding head being out of spec. How would you tell the two apart using only this station's dashboard, without opening the tool?*

Key teaching point: the same spatial shape can come from two very different root causes (mechanical damage vs. one bad equipment head). The shape narrows your search — it does not replace checking this station's own parameter readings.

Wrap-Up

Instructor: *Today's Wire process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between a ball bond and a wedge bond?

A. A ball bond forms a ball via EFO before thermocompression (typically the 1st bond); a wedge bond compresses directly without a ball.

Q. Why does Loop Height matter for downstream reliability?

A. Too low risks shorting from wire sag; too high risks wire sweep during molding.

II-3. [Session 3] Molding

Learning Objectives

- Explain the principles and reliability-management points of the Molding process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

Instructor: *Today's process encapsulates the die and wires in Epoxy Molding Compound (EMC). Mold-fill control across the preheat-cure-cool profile is central.*

Development — Simulator-Linked Instruction

- On the MODULE 03 screen, play the sequence where the mold clamps and EMC resin fills the cavity under transfer pressure.
- Explain the three-stage temperature profile (preheat → cure → cool), and how PID overshoot causes premature EMC curing and incomplete fill.
- Learner activity: adjust transfer-pressure and cure-time sliders and observe how Defect Rate responds when values fall outside the target windows (pressure 70 ± 8 kgf/cm², cure time 90 ± 15 sec).

PROCESS ANOMALY SCENARIO — Low-Pressure Fill Causing Incomplete Runner Fill / Void

Root cause: Transfer pressure drops below spec, so the mold cavity does not fully fill with EMC and voids form.

Symptom (dashboard reaction): Defect Rate rises; Yield Trend shifts to a downward slope.

Instructor action: Have learners readjust the transfer-pressure parameter and demonstrate the fill-quality recovery.

6D FAULT CASE — Expected Pattern: Center

Multi-cavity molds are standard in production. Center cavities of a mold platen tend to run hotter than edge cavities due to heat concentration, so if the cure profile drifts, center-cavity units cure differently from edge-cavity units.

Instructor: *If your mold has 16 cavities arranged in a grid and the center cavities are consistently a few degrees hotter, what shape would you expect on the map once those units are laid out in their original tray positions?*

This is the back-end mirror of front-end's Center pattern (chuck hot-spot) — same shape, same underlying idea (a hot zone in the middle of a fixture), completely different physical stage of the flow.

Wrap-Up

Instructor: *Today's Molding process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. Why is minimizing PID overshoot important?

A. If cure temperature exceeds target, the EMC cures prematurely, causing fill defects and elevated internal stress.

Q. What is the primary cause of package warpage?

A. CTE (coefficient of thermal expansion) mismatch between the EMC and the leadframe/substrate causes uneven shrinkage on cooling.

II-4. [Session 4] Saw / Dicing

Learning Objectives

- Explain the principles and reliability-management points of the Saw process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

Instructor: *Today's process cuts the molded strip into individual packages with a diamond blade. Blade-wear management and SFC (equipment controller) design are central.*

Development — Simulator-Linked Instruction

- On the MODULE 04 screen, play the sequence of the high-speed rotating blade cutting the strip while coolant sprays.
- Explain that as a blade wears, cutting resistance rises, rotation speed becomes unstable, and micro-chipping appears at die edges.
- Learner activity: adjust blade-rotation-speed and coolant-flow sliders and observe how chipping rate (Defect Rate) responds. Introduce how SFC (Smart Factory Controller) design and EtherCAT communication let a real line monitor blade condition in real time.

⚠ PROCESS ANOMALY SCENARIO — Blade Wear Causing Chipping Defects

Root cause: Progressive blade wear raises cutting resistance, pushing rotation speed out of spec and causing micro-chipping at die edges.

Symptom (dashboard reaction): Defect Rate rises; coolant flow also drops below spec at the same time.

Instructor action: Have learners readjust rotation-speed and coolant-flow parameters and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Scratch (Line)

A dicing saw cuts along fixed streets (lanes) across the strip. If one specific street develops a problem — a locally worn section of blade edge, a resin buildup, or a kerf-width deviation on that lane — every package along that one street shows chipping, forming a literal straight line.

Instructor: *Notice this is the second module in a row that can produce a Scratch pattern — but for a completely different physical reason than Wire Bonding's. Here, the line runs along a physical cutting path; at Wire Bonding, it followed an equipment-head assignment. How does knowing which module you're looking at change how you'd confirm the cause?*

Use this back-to-back Scratch example deliberately: it proves the point from Session 2 that identical map shapes require checking the originating station's own dashboard data to fully confirm the mechanism, not just the shape.

Wrap-Up

Instructor: *Today's Saw process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between blade dicing and stealth dicing?

A. Blade dicing cuts physically; stealth dicing forms an internal modified layer with a laser, then separates mechanically with less chipping.

Q. What happens when coolant flow is insufficient?

A. Heat builds up on the blade, accelerating wear and increasing both chipping rate and thermal damage.

II-5. [Session 5] Package Testing

Learning Objectives

- Explain the principles and reliability-management points of the Package process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

Instructor: *Today is the final back-end gate: inspecting the finished package's appearance and electrical characteristics. AOI, X-ray, Pull Test, and 6D spatial pattern reading are all central here.*

Development — Simulator-Linked Instruction

- On the MODULE 05 screen, play the sequence of a socket contacting an individual package to measure electrical signals while an AOI camera scans.
- Explain how the AOI decision threshold is set and periodically recalibrated, and how a threshold set too loosely lets genuinely defective units pass as good (a false accept).
- Learner activity: use the new Package Map Pattern Drill panel to practice reading all six spatial patterns (Edge · Center · Donut · Scratch · Random · Local Cluster) hands-on — use alongside the companion Wafer Map Pattern Recognition Reference Card and Scripted Guide.

PROCESS ANOMALY SCENARIO — AOI Threshold Drift Causing False Accepts

Root cause: Illumination degradation shifts the AOI decision threshold outside the 50 ± 8 spec, causing genuinely defective packages to pass as good.

Symptom (dashboard reaction): Yield reads higher than reality, while the underlying Defect Rate data is quietly distorted.

Instructor action: Demonstrate readjusting the AOI-threshold parameter to restore inspection reliability. Use this scenario to emphasize the difference between 'an error in the inspection itself' and 'an actual process defect.'

◆ 6D FAULT CASE — Expected Pattern: Random (Scatter)

An AOI threshold drift is a calibration problem, not a location-based defect — it doesn't care where on the tray a unit sits. Any unit, regardless of position, has an equal chance of being misjudged, so the map shows no shape at all.

Instructor: *This is the most important fault case in the whole back-end series, because it's the one where 6D correctly tells you there IS no spatial story. If a team sees a random scatter and starts hunting for a station with a location-based cause, what should you tell them?*

This deliberately completes the set: four of the five back-end modules can produce a shape (Cluster, two different Scratch causes, Center); this one produces no shape at all — and recognizing 'no shape' as diagnostic information is itself an advanced 6D skill, not a failure to find one.

Wrap-Up

Instructor: *Today's Package process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. Why must the false accept rate stay low?

A. A defective unit shipped as good leads to field failure (RMA), which carries far higher cost and reliability risk than catching it at test.

Q. Why does the 6D reading at Package Testing matter?

A. A number alone (Defect Rate) can't reveal the cause, but the spatial shape of a package map (e.g., a straight-line scratch) immediately narrows the candidate causes — including, as this session shows, telling you when there is no spatial cause at all.

II-6. [Session 6] Integration · 5-Process Chain Scenario & Certification

Learning Objectives

- Operate all five back-end modules as a single, continuous digital-twin scenario.
- Trace a final Package Testing result back to the originating process stage — root-cause tracing.
- Verify learning outcomes through a certification assessment that includes 6D spatial pattern reading.
- Use the 6D Fault Case Gallery to recognize that identical shapes can come from different stations and different mechanisms.

Intro

Instructor: *Today we connect everything you learned over the past five weeks into a single package journey. You now have both the expertise of each individual process and the perspective of a reliability engineer surveying the entire line.*

Development — Full-Scenario Practice

- Run the Full Digital Twin Dashboard to operate the integrated simulation where all five stages, Die Bonding through Package Testing, are sequentially linked.
- The instructor secretly inserts an anomaly condition into one of the five processes beforehand. Learners see only the final Package Testing result — both the numeric data and the spatial pattern — and trace back the originating process in teams (40-minute diagnostic mission).
- Each team presents its diagnosis and evidence (numeric data + pattern shape); the instructor compares it against the true origin and gives feedback.

6D Fault Case Gallery — Reference

Review this table with learners before the diagnostic mission. It summarizes every fault case covered this program, including the deliberate overlap between Wire Bonding and Saw/Dicing.

Module	Fault	Expected 6D Pattern	Why
Die Bonding	Under-dispense void	Local Cluster	Nozzle clog degrades gradually — a run of consecutive units is affected, not the whole tray.
Wire Bonding	Loop height / EFO drift	Scratch (Line)	One bonding head, fixed to specific tray positions, drifts out of spec while others stay fine.
Molding	Low-pressure incomplete fill	Center	Center cavities of a multi-cavity mold run hotter, curing differently from edge cavities.
Saw / Dicing	Blade wear chipping	Scratch (Line)	One dicing street/lane develops a local blade issue — every unit along that lane is affected.
Package Testing	AOI threshold drift	Random (no shape)	A calibration problem has no location dependence — any unit can be equally affected.

Discussion: two rows share the same 'Scratch' pattern. Ask learners how they would distinguish a Wire Bonding cause from a Saw/Dicing cause using only the shape — the honest answer is that they can't, until they check each station's own dashboard readings.

Certification Assessment

- Written assessment: 15 questions covering core concepts from every session plus 6D pattern reading (20 min).
- Practical assessment: diagnose and respond to one anomaly scenario in the simulator, plus read one spatial pattern correctly (25 min).

- Passing criteria: 70+ on the written assessment, and 5 of 6 items met on the practical checklist, to receive a certificate of completion.

Wrap-Up

Instructor: *As of today, you can diagnose the entire journey from a single die to a finished package.*

Combined with the front-end program, you're now ready to take on the full 13-station Wafer to Package Capstone. Well done.

Checkpoint Q&A

Q. Why does root-cause tracing matter across a 5-process chain?

A. A final defect can result from accumulated deviations across multiple steps, so the full process history — and the spatial evidence — must be examined rather than just the final test result.

Q. Why include 6D pattern reading in the certification assessment?

A. Number-based diagnosis and spatial-pattern-based diagnosis require different reasoning processes, and real yield-review meetings on the shop floor require both abilities together.